

CS345: Algorithms II

Semester I, 2026–27

29 July 2026

Course Information

Instructor	Raghunath Tewari
Office	Room 514, RM Building
Email	rtewari@cse.iitk.ac.in (preferred communication mode)
Lectures	Monday, Wednesday, Friday, 9:00–10:00 AM, RM101
Course Website	Moodle (moodle.cse.iitk.ac.in)
Prerequisite	ESO207: Data Structures and Algorithms

Course Objectives

This course studies fundamental algorithmic techniques beyond the introductory level. By the end of the course, students should be able to

- design efficient algorithms for a wide variety of computational problems,
- prove correctness using rigorous mathematical arguments,
- analyse running time and memory usage,
- compare different algorithmic paradigms and identify the appropriate one for a given problem, and
- understand the computational complexity of algorithmic problems.

Course Syllabus

1. Divide-and-conquer algorithms and advanced applications.
2. Binary search trees, interval trees, and orthogonal range searching.
3. Greedy algorithms and applications.
4. Graph algorithms:

- Dijkstra’s algorithm
 - Bellman–Ford algorithm
 - Floyd–Warshall algorithm
 - Advanced applications of DFS
5. Dynamic programming.
 6. Network flow:
 - Ford–Fulkerson algorithm
 - Max-flow min-cut theorem
 - Applications
 7. Amortized analysis.
 8. String matching:
 - Knuth–Morris–Pratt algorithm
 9. Binomial heaps and Fibonacci heaps.
 10. Selection algorithms and finding the median using small space.
 11. Stable matching (Gale–Shapley algorithm).
 12. Polynomial-time reductions.
 13. NP, NP-completeness, and reductions.
 14. Introduction to approximation algorithms.

Recommended References

- Cormen, Leiserson, Rivest and Stein, *Introduction to Algorithms*, 4th Edition.
- Kleinberg and Tardos, *Algorithm Design*.

Learning Outcomes

Upon successful completion of the course, students will be able to

- design efficient algorithms using divide-and-conquer, greedy, and dynamic programming techniques;
- analyse the correctness and complexity of algorithms;
- solve graph problems using shortest-path, network flow, and matching algorithms;
- apply advanced data structures to algorithm design;
- understand the notion of computational intractability through NP-completeness; and
- appreciate the role of approximation algorithms in solving hard optimization problems.

Testing and Grading

Grading will be based on attendance, homework assignments, quizzes, a mid semester exam and a final exam. The following table gives a guideline for evaluating your final grade.

Component	Weight
Assignments	5%
Quizzes	25%
Mid-semester Examination	30%
End-semester Examination	40%
Total	100%

Other Important Information

- There will be two graded assignments. Submissions must be made through Moodle before the specified deadline. Late submissions will incur a penalty unless prior permission has been obtained.

Additionally, two or three ungraded assignments will be provided for practice. Although these assignments do not contribute to the final grade, students are strongly encouraged to solve them independently.

- There will be two to four quizzes during the semester. Some quizzes may be announced while others may be conducted without prior notice.
- Make-up examinations or quizzes will be conducted only under exceptional circumstances and solely at the discretion of the instructor.
- Academic honesty is expected from every student. Any form of plagiarism, copying, unauthorized collaboration, impersonation, or other academic misconduct will be treated as a serious violation of the Institute's academic integrity policy and may result in severe penalties.
- Clarity and legibility of your solutions are as important as the solution itself.
- All course information will be conveyed via the course mailing list. Please make sure you check your email regularly.